

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Second Semester of M. Tech (IT) Examination****May 2015****IT720 Advanced Operating System****Date: 11.05.2015, Monday****Time: 1:30 p.m. To 4:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the below given questions:****[07]**

- a) In most cases, if a process is sent a signal while it is executing a system call :
A. the system call will continue execution and the signal will be ignored completely.
B. the system call is interrupted by the signal, and the signal handler comes in.
C. the signal has no effect until the system call completes.
D. None of these.
- b) Which signal is sent when the Child process terminates?
A. SIGINIT **B.** SIGKILL **C.** SIGSTOP **D.** SIGCHLD
- c) Suppose the value of umask is 0002. A new file called "AOS" is created . What are the default permissions for this newly created file?
A. 665 **B.** 664 **C.** 775 **D.** 774
- d) Microprocessor was introduced in which generation of Computer?
A. Second Generation **B.** Third Generation **C.** Fourth Generation **D.** Both B and C
- e) State true or false: Named Pipe has entry in the directory.
- f) Which directory contains configuration files that stores system and application setting?
A. /etc **B.** lib **C.** /usr **D.** /var
- g) How can you erase all files in the current directory, including all its sub-directories, using only one command?

Q – 2.a Briefly explain the difference between the user mode and kernel mode operations. Why two modes of operations are required in UNIX? **[04]**

Q – 2.b What do you mean by block read ahead? How it can be implemented? **[06]**

OR

Q – 2.b How is I/O by polling different from I/O by interrupt? **[06]**

Q – 2.c List and explain the major sources of signal generation. [04]

OR

Q – 2.c What resources are used when a thread is created in Mach operating system? How do they differ from those when a process is created? [04]

Q - 3 Answer the following:(Any two) [14]

- Describe the Kernel Architecture of Unix Operating System.
- Elaborate on Scenarios for the retrieval of a buffer.
- Enlist the state information that are to be maintained in each user thread with one line explanation. What do you mean by thread scheduling?

SECTION – II

Q - 4 Do as directed: [06]

- How do you create special files like named pipes ? [01]
- What is the task of psig()? [01]
- List and explain in one line, two system calls used for process management. [02]
- What is the problem caused by file locking? How it can be resolved? [02]

Q – 5.a Consider the given double linked list(figure a). A new node called bp1 is to be inserted in the list through the code given in figure b. Suppose a context switch happens when the code reaches the comment. What would be the status of linked list before process completes executing code before the comment? What is the solution to the problem? [05]

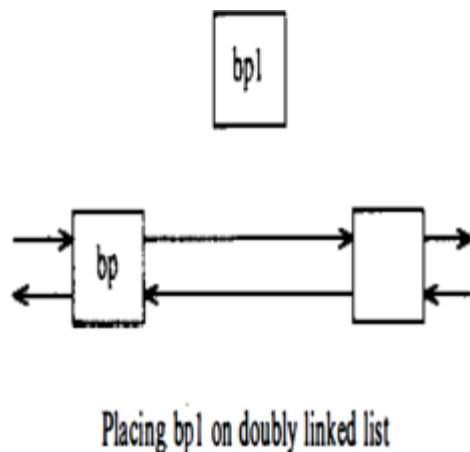


Figure a

```
struct queue {  
    ) *bp, *bp1 ;  
    bp1 -> forp = bp- > forp;  
    bp1 -> backp = bp;  
    bp- > forp = bp1 ;  
    /*consider possible context switch  
    here */  
    bp1 -> forp- > backp = bp1 ;
```

Figure b.

Q – 5.b Describe all the conditions where reference count of an inode can be greater than 1. **[05]**

OR

Q – 5.b How do you implement a mount system call? **[05]**

Q – 5.c List and describe the data structures used by kernel in brief. **[05]**

OR

Q – 5.c List and explain different multithreading models and compare them. **[05]**

Q – 6 Attempt any Two: **[14]**

- a. Draw the architecture of Satellite processor distributed system. How to handle signals on satellite systems.
- b. What is Light weight process? List and explain several factors that influence the design of lightweight process.
- c. Explain lseek system call with its arguments and example.
